

# Khawaja Yunus Ali University

School of Science and Engineering

Department of Computer Science and Engineering

Final Examination: Fall Semester 2019

Program: B.Sc. in CSE (Batch: 9<sup>th</sup>) 1<sup>st</sup> Year 1<sup>st</sup> Semester

Course Title: Physics – I

Course Code: PHY 1101

Total Time: 2 hours

Full Marks: 40

## Part-B

[Answer the question No. 1 and any four (04) from the rest]

5 × 6 = 30

1. Read the stem given below and answer the questions that follows –

A small search coil with an area of  $125\text{mm}^2$  has 50 turn of very fine wire. This coil is placed between the pole pieces of a small magnet and then suddenly jerked out. If the average induced emf is  $0.07\text{V}$  when the coil is pulled to a field -free region in  $60\text{ms}$ .

- |    |                                                                                                                                                                                                                    |   |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| a) | Define 1 henry.                                                                                                                                                                                                    | 1 |
| b) | What is the magnetic intensity between the poles?                                                                                                                                                                  | 3 |
| c) | Investigate the original flux through each turn.                                                                                                                                                                   | 2 |
|    |                                                                                                                                                                                                                    |   |
| 2. | a) Can a transformer be used with DC? – justify your answer.                                                                                                                                                       | 3 |
|    | b) How does lenz law follow conservation of energy?                                                                                                                                                                | 3 |
|    |                                                                                                                                                                                                                    |   |
| 3. | a) Write down the differential equation of simple harmonic motion (SHM).                                                                                                                                           | 1 |
|    | b) Mention the conditions for simple harmonic motion (SHM).                                                                                                                                                        | 2 |
|    | c) Find the amplitude, frequency & time period of SHM equation:<br>$y = 50\sin(628t + \delta)$                                                                                                                     | 3 |
|    |                                                                                                                                                                                                                    |   |
| 4. | a) Define the reduced mass of two body oscillation.                                                                                                                                                                | 2 |
|    | b) If two bodies of masses 5g and 10g are connected by a massless spring of force constant $1.5\text{dynes/cm}$ , then calculate the effective mass and time period of oscillation.                                | 4 |
|    |                                                                                                                                                                                                                    |   |
| 5. | a) Define 1 Tesla.                                                                                                                                                                                                 | 1 |
|    | b) An electron is moving with a speed of $5 \times 10^7\text{m/s}$ at right angle to a magnetic field of 5000G. (a) What is the magnetic force on the electron?                                                    | 2 |
|    | (b) What is the radius of circle in which the electron moves?                                                                                                                                                      | 3 |
|    |                                                                                                                                                                                                                    |   |
| 6. | a) State Biot-Savart law. Write down the vector form of it.                                                                                                                                                        | 2 |
|    | b) Draw a figure which can create torque in a current carrying coil in the magnetic field.                                                                                                                         | 1 |
|    | c) A coil of 20 turns has an area of $800\text{mm}^2$ and bears a current of 0.5A. It is placed with its plane parallel to magnetic field of intensity 0.3T. Determine the torque and magnetic moment on the coil. | 3 |

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## Part-A

Time: 20min

[Tick (✓) mark to the appropriate one]

0.5 × 20 = 10

1. The periodic time (T) is given by .....
  - a)  $\omega/2\pi$
  - b)  $\omega/2$
  - c)  $\omega/\pi$
  - d)  $2\pi/\omega$
  - e)  $\pi/\omega$
2. The maximum displacement of a body, from its mean position is called .....
  - a) amplitude
  - b) time period
  - c) frequency
  - d) wave length
  - e) acceleration
3. Magnitude of the magnetic field is ..... to the strength of current.
  - a) proportional
  - b) equal
  - c) inversely proportional
  - d) independent
  - e) less than
4. The transformer used in .....
  - a) DC
  - b) AC
  - c) DC battery
  - d) zero voltage
  - e) capacitor
5. What is the maximum energy of simple harmonic motion?
  - a)  $\frac{1}{2}KA^2$
  - b)  $\frac{1}{2}KA$
  - c)  $KA^2$
  - d)  $KA$
  - e)  $\frac{1}{2}K$
6. What is the relation between magnetic field and magnetic flux?
  - a)  $B = \frac{\phi}{2}A$
  - b)  $B = \frac{i}{\phi}A$
  - c)  $B = \frac{A}{\phi}$
  - d)  $B = \frac{\phi}{A}$
  - e)  $B = A\phi$

7. What is the unit of the magnetic flux?
- $T$
  - $wb$
  - $Gauss$
  - $Volt$
  - $A$
8. How many emf generated when a coil of 100 turns is subjected to a flux rate of 0.3 tesla/sec?
- 0.003
  - 0.03
  - 0.3
  - 3
  - 30
9. According to the Ampere's law,  $\oint \vec{B} \cdot d\vec{l} = ?$
- $i$
  - $\frac{i}{\mu_0}$
  - $\mu_0 i$
  - $\frac{\mu_0}{i}$
  - $\mu_0$
10. What is the unit of hall voltage?
- $Nm$
  - $N$
  - $volt$
  - $A$
  - $Am^2$
11. The magnetic field of a current carrying wire in the inside of itself is -
- $10^{-7} TmA^{-1}$
  - $4\pi TmA^{-1}$
  - $4\pi \times 10^{-7} TmA^{-1}$
  - $\pi$
  - zero
12. Two parallel wire carrying current in opposite direction, .....
- independent to each other.
  - join each other.
  - attract each other.
  - repel each other.
  - burn each other.
13. Current carrying solenoid act as an -
- toroid
  - magnet
  - insulator
  - semiconductor
  - source
14. The number of charges per unit volume of a conductor can be determined by -
- Biot-savart's law
  - Ampere's law
  - Hall effect
  - Lorenz force
  - Lenz's law

15. What is the frequency of home electric supply line in Bangladesh?
- a)  $50Hz$
  - b)  $65Hz$
  - c)  $75Hz$
  - d)  $85Hz$
  - e)  $100Hz$
16. Correct expression for the magnetic field inside a solenoid along the axis .....
- a)  $B = \mu_0 IM$
  - b)  $B = \mu_0 n$
  - c)  $B = \mu_0 ni$
  - d)  $B = \mu_0 \phi$
  - e)  $B = \mu_0 \phi i$
17. What is the unit of the e.m.f.?
- a)  $T$
  - b)  $wb$
  - c)  $Gauss$
  - d)  $Volt$
  - e)  $A$
18. The value of proportionality constant of Newman's law is –
- a) 1
  - b)  $4\pi$
  - c)  $\pi$
  - d)  $10^{-10}$
  - e)  $10^{-7}$
19. Which one is the right relationship between angular displacement and angular frequency?
- a)  $\theta = \omega t$
  - b)  $\theta = \omega n$
  - c)  $\theta = \omega \pi$
  - d)  $\theta = \pi n$
  - e)  $\theta = \sigma t$
20. The outward direction of magnetic field in a plane paper is indicated by -
- a) alpha sign
  - b) beta sign
  - c) tick sign
  - d) cross sign
  - e) dot sign

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Invigilator Signature

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Examiner Signature